

HOUSE PRICE PREDICTION – WEEK 1 PROJECT SUMMARY

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Internship: XYlofy AI Internship Program

Project Title: House Price Prediction

Tools Used: Python, Pandas, Scikit-learn, Matplotlib, Seaborn, Jupyter Notebook

PROJECT OBJECTIVE

The objective of this project was to develop a machine learning model capable of predicting residential house prices using property-related features such as area, number of bedrooms and bathrooms, parking availability, furnishing status, and other amenities. The project demonstrated the end-to-end workflow of data analysis, preprocessing, model training, evaluation, and visualization using Python.

DATASET

The Housing Prices Dataset sourced from Kaggle was used. It contains multiple attributes related to residential properties, with price as the target variable representing the selling price of each property.

WORK PERFORMED

- Loaded and explored the dataset using Pandas; checked shape, columns, and missing values
 - Removed duplicate records and converted categorical variables to numerical format using one-hot encoding
 - Split the dataset into training and testing sets (80:20 ratio)
 - Trained two machine learning models: Linear Regression and Random Forest Regressor
 - Evaluated model performance using MAE, RMSE, and R^2 Score
 - Created visualizations: price distribution histogram, correlation heatmap, and Actual vs. Predicted scatter plot
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KEY FINDINGS

Features such as area, number of bathrooms, number of stories, parking availability, and air conditioning showed the strongest influence on house prices. Larger properties with better amenities consistently commanded higher prices. The correlation heatmap confirmed positive relationships between these features and the target variable.

MODEL PERFORMANCE

The Random Forest Regressor outperformed Linear Regression by capturing complex, non-linear relationships between features and price. It produced lower prediction errors (MAE and RMSE) and a higher R^2 score, indicating superior overall accuracy.

CONCLUSION

This project successfully demonstrated the application of machine learning to real estate price prediction. Through data cleaning, feature encoding, regression modeling, and performance evaluation, meaningful insights were extracted from the dataset. The results highlight the potential of ML-driven tools to support accurate property valuation and data-driven decision-making in the real estate industry. The project also provided hands-on experience with the complete data science pipeline in Python.